

Object-Oriented Python

Vassilis Markos

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E1. Consider the following two python classes:

```
1 class A:
2     def __init__(self, n):
3         self.n = n
4
5     def __eq__(self, __other):
6         if not isinstance(__other, A):
7             return False
8         return self.n == __other.n
9
10 class B(A):
11     def __init__(self, n):
12         self.n = n
13
```

- (a) Explain what the above code snippet defines. How are the two classes defined? Is there any sort of inheritance? Are we overwriting any default python object methods?
- (b) What will the following python script print?

```
1 if __name__ == "__main__":
2     x = A(1)
3     y = A(2)
4     print(f"Are x and y equal? {x == y}")
5
```

- (c) What about the following?

```
1 if __name__ == "__main__":
2     x = A(1)
3     y = A(1)
4     print(f"Are x and y equal? {x == y}")
5
```

- (d) What about the following?

```
1 if __name__ == "__main__":
2     u = B(1)
3     v = B(1)
4     print(f"Are u and v equal? {u == v}")
5
```

(e) What about the following?

```
1 if __name__ == "__main__":
2     u = B(1)
3     v = A(1)
4     print(f"Are u and v equal? {u == v}")
5
```

(f) What if we define the two classes as follows? How do the above cases change, if at all?

```
1 class A:
2     def __init__(self, n):
3         self.n = n
4
5 class B(A):
6     def __init__(self, n):
7         self.n = n
8
9     def __eq__(self, __other):
10        if not isinstance(__other, A):
11            return False
12        return self.n == __other.n
13
```

E2. In this exercise we will create a (really–really) simple movie database management system.

(a) As a first step, create a `Movie` class, with the following fields:

- **name**, a string representing the name of the movie;
- **genres**, a list of strings indicating the genre(s) of the movie;
- **duration**, an integer indicating the duration of the movie in minutes;
- **director**, a string containing the name of the movie director;
- **cast**, a list of strings, containing the names of the movie cast.

Moreover, the class should overwrite python's `__str__` method so as to print a movie as follows (with tabs, spacing and everything as shown below):

```
Movie: A Christmas Carol
      Director: Robert Zemeckis
      Genre   : Animation, Adventure, Comedy
      Duration: 126 minutes
      Cast    : Jim Carrey, Garry Oldman, Colin Firth
```

You can read more about python's `__str__` magic function here:
https://docs.python.org/3/reference/datamodel.html#object.__str__

- (b) Add an object method, `write_to_file()`, to class `Movie` that accepts a filename as an argument and adds a line to that file with all movie information in the following form:

```
name | director | genre | duration | cast
```

where `|` is used as a delimiter, to distinguish between the various fields. For instance, for the movie “A Christmas Carol” shown above, this line should be as follows (split here just to fit in the page):

```
A Christmas Carol | Rober Zemeckis | Animation,  
Adventure, Comedy | 126 | Jim Carrey, Garry Oldman,  
Colin Firth
```

- (c) Add an object method, `from_text()`, that accepts a string of the following form:

```
name | director | genre | duration | cast
```

and updates all class fields accordingly.

If you wish, you can first read about how this can be achieved by using class methods¹ but it is not compulsory for this part of the exercise to use python’s `@classmethod`.

- (d) Write a python script that:

- Asks the user to provide the name of a movie and then asks for additional information:
 - movie director,
 - genres,
 - duration,
 - cast.
- It creates a `Movie` object with the provided values and writes it to a file (the file has to be created for the first time this script is used).
- It repeats the above until the user enters “:q” as the name of the movie.

- (e) Consider extending the above python script by adding several functionalities. These could include (but should not be restricted to) the following:

- duplicate checks, i.e. check whether a movie has already been written into the file;
- querying a movie, i.e., allowing the user not only to enter a movie but also ask for information about a movie;

¹One related source: <https://stackoverflow.com/questions/5249813/create-python-objects-from-strings>.

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Try to provide well-thought implementations. For instance, if you implement a querying functionality, what if the movie is not included in the file? How are you going to actually search the file?